

3(a)	positions of H and OH for C1 and C4 are other way round / AW ; can award from a diagram A H should be up on C1 and C4 A OH should be down on C1 and C4	1
3(b)	<i>conclusion 1 there are non-reducing and reducing sugars in the carrots</i> (all have) fructose/galactose/glucose, which, is / are, reducing (sugar(s)) and sucrose, which is <u>non-reducing</u> (sugar) ; A sucrose is non-reducing and all the others are reducing (sugars) A fructose / galactose / glucose, reacts with Benedict's solution and sucrose does not R if another sugar in addition to sucrose is stated as non-reducing <i>conclusion 2 there are monosaccharide and disaccharide sugars in the carrots</i> (all have) fructose / galactose / glucose, which, is / are, monosaccharide(s) and sucrose, which is a disaccharide ; R if another sugar in addition to sucrose is stated as disaccharide <i>conclusion 3 the carrots have the same patterns of results</i> (yes) all have order of content as (highest) fructose, glucose, sucrose and galactose ; A all have fructose as highest content and galactose as lowest content I data quotes	3
3(c)	starch / amylose / amylopectin / cellulose / hemicellulose / pectin / inulin ;	1

Question	Answer	Marks
3(d)	<i>allow other AW for assimilates e.g. sugars / sucrose / other named sugars / amino acids / organic compounds</i> <i>allow named locations for growing areas e.g. developing shoots / developing leaves / buds / seeds</i> <i>ignore 'nutrients' as AW for assimilates but allow ecf</i> acts as source (only), after dormant period/in second year of growth <u>and</u> is a sink in the first year of growth ; I during dormant period one detail of source ; e.g. <i>idea that</i> assimilates are mobilised for transfer e.g. starch to sugars for transport assimilates, loaded / AW, into phloem (tissue) source is starting area of, translocation / transport of assimilates source, provides assimilates for growing areas one detail of sink ; e.g. of, growth / development storage e.g. energy store / assimilates converted for storage / storage of substances receives assimilates transported, in phloem sap / in phloem tissue / photosynthates receives assimilates from, source / leaves / sites of synthesis / photosynthetic tissue	3
3(e)	<i>any two from:</i> has RNA (as genetic material), not DNA / so does not need transcribing / so transcription does not occur ; (viral) RNA, acts as mRNA / is sequence of mRNA codons ; AW (viral) RNA can be used (directly) for translation ; A description of translation e.g. to form, amino acid chain / polypeptide I for synthesis of proteins attach to ribosomes and make, polypeptide / protein <i>ref. to single— / double—, stranded is neutral</i>	2